

Closing the loop from pathogenic variant to gene therapy

Scaling our ability to explore the pareto frontier of patient-specific mechanistic hypotheses and therapeutic variants

About Vivek

- > 20 years in drug discovery @ Genentech
 - 10 years focused on computational research biology (what causes disease?)
 - 10 years focused on computational antibody engineering (how do we treat diseases?)

Pareto frontiers lie at the center of drug discovery

Find solutions
that balance
competing
objectives

Mechanistic hypotheses

Strength of the evidence vs **Druggability?**

Biological therapeutics

Efficacious and on target vs **avoid immune system**

A central fight in drug discovery

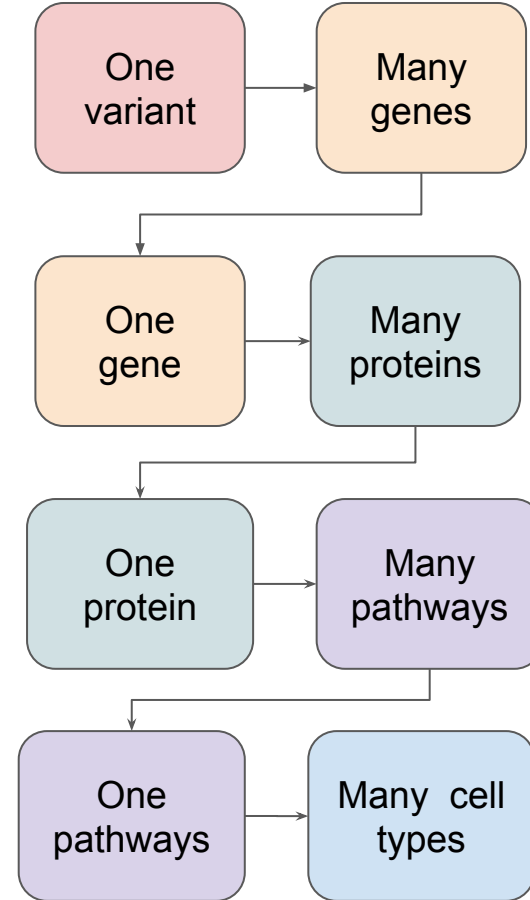
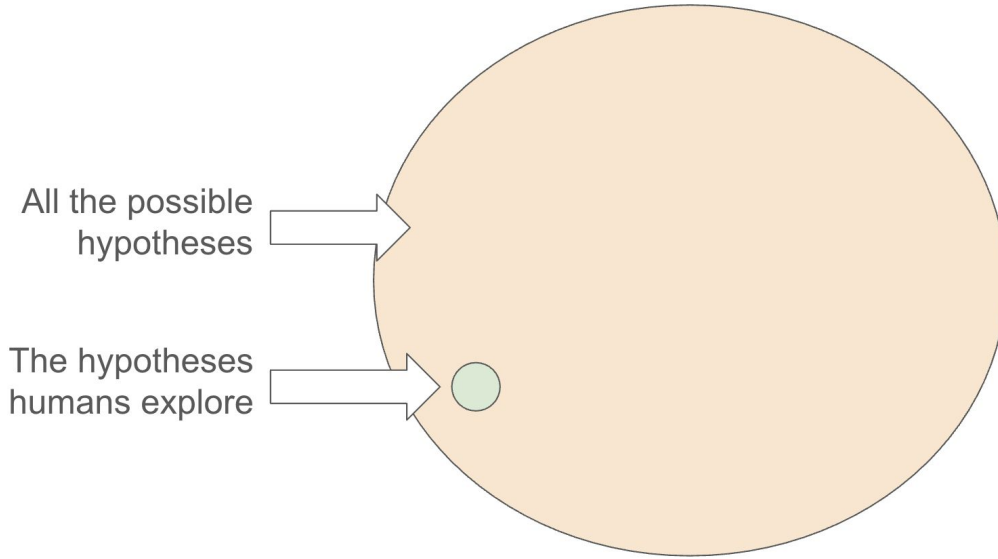
Identifying solutions in the
pareto frontier

Human
expertise is
critical

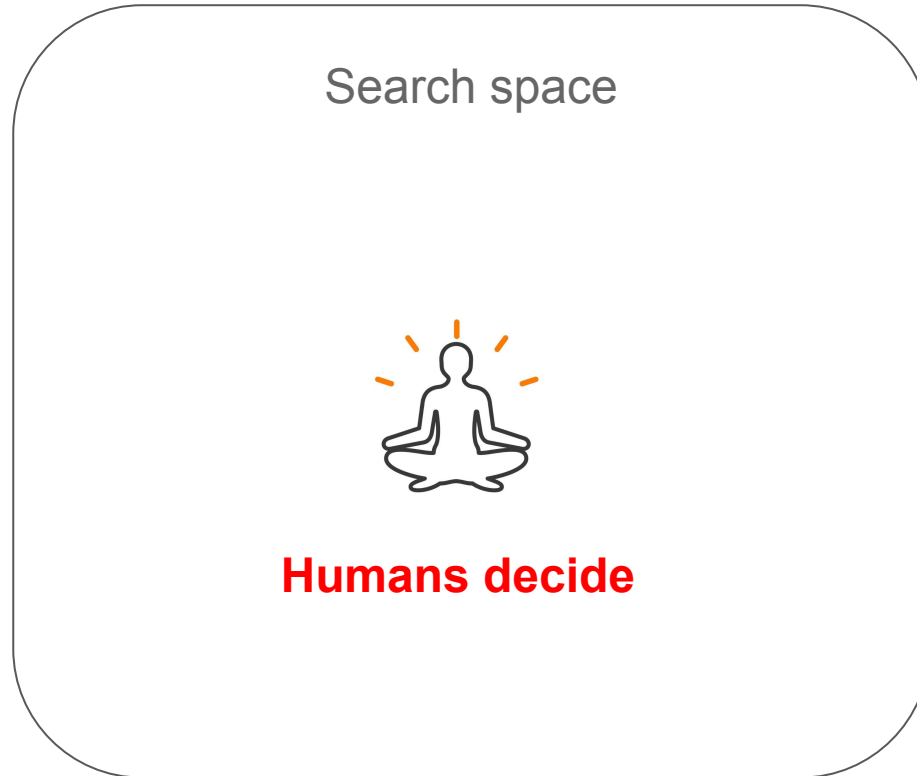


Human
expertise is a
form of bias

The basis for this fight



The answer is attention



Machines guide

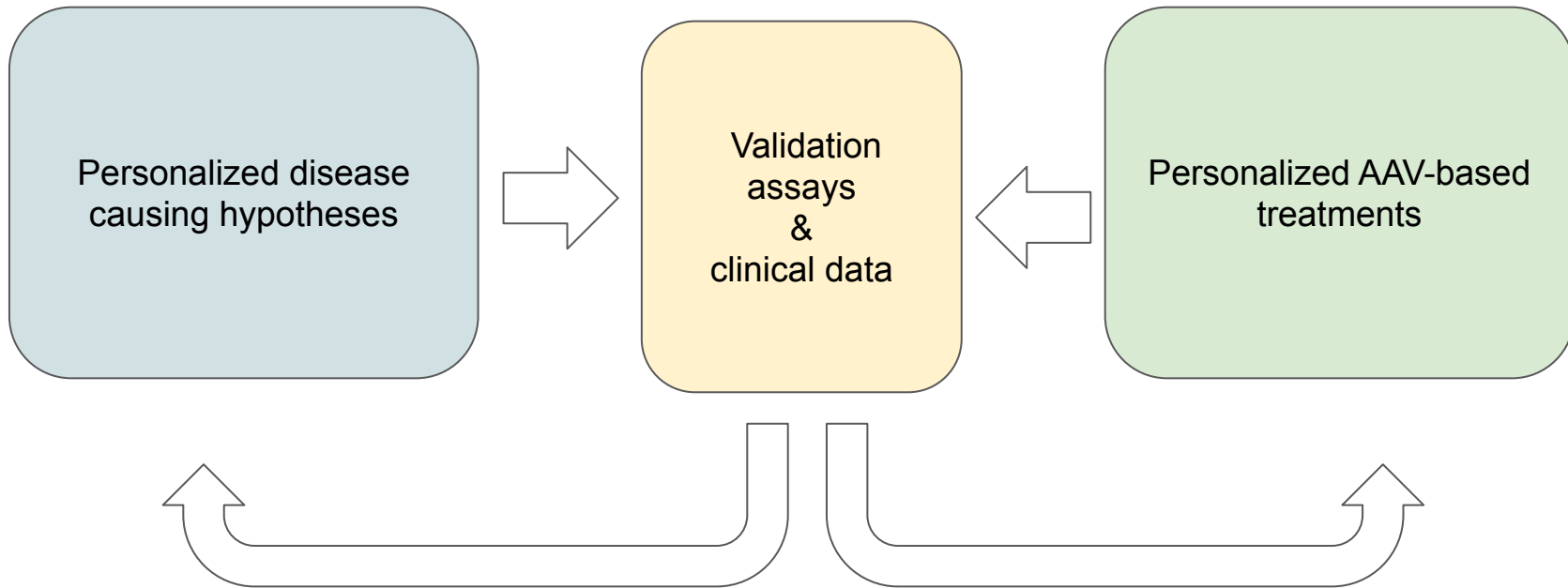
Rare disease as a test bed

Highly motivated
patient and
caregiver
populations

Limited numbers
of clinicians and
researchers

Few, if any,
therapeutic
options

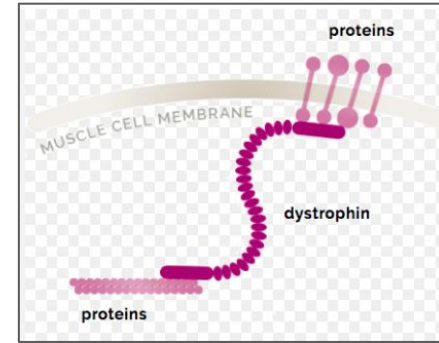
I'm building a platform with rare disease researchers



Today: DMD (Duchenne's Muscular Dystrophy)

What is the mechanism by which pathogenic variants result in observed phenotypes?

Which AAV therapeutics are optimized to treat this mechanism of disease?



Primary Organ Systems Affected by DMD

- **Musculoskeletal System:** Progressive loss of skeletal and postural muscles leads to early childhood weakness, loss of walking ability, and spinal curves like scoliosis. [Rare Disease Advisor +4](#)
- **Cardiovascular System:** Heart muscle damage causes progressive dilated cardiomyopathy and heart rhythm issues (arrhythmias). [Rare Disease Advisor +3](#)
- **Respiratory System:** Weakness in the diaphragm and chest wall muscles leads to declining lung volume, ineffective coughing, and eventual respiratory failure. [Rare Disease Advisor +2](#)
- **Nervous System & Brain:** Dystrophin isoforms in the brain can lead to neurodevelopmental variations, learning differences, or cognitive and behavioral challenges. [Rare Disease Advisor +3](#)
- **Gastrointestinal System:** Smooth muscle weakness of the gut causes swallowing difficulties, delayed stomach emptying, and chronic constipation. [Rare Disease Advisor +2](#)
- **Endocrine & Skeletal (Bone) Systems:** Long-term corticosteroid treatments and immobility contribute to slowed growth, delayed puberty, and fragile bones (osteoporosis/fractures). [Rare Disease Advisor +2](#)

Demo video

Pareto frontier — therapeutics

212 capsid variants · 4 on Pareto frontier · constraint: hepatotoxicity < 0.40

STRATEGY:

all strategies

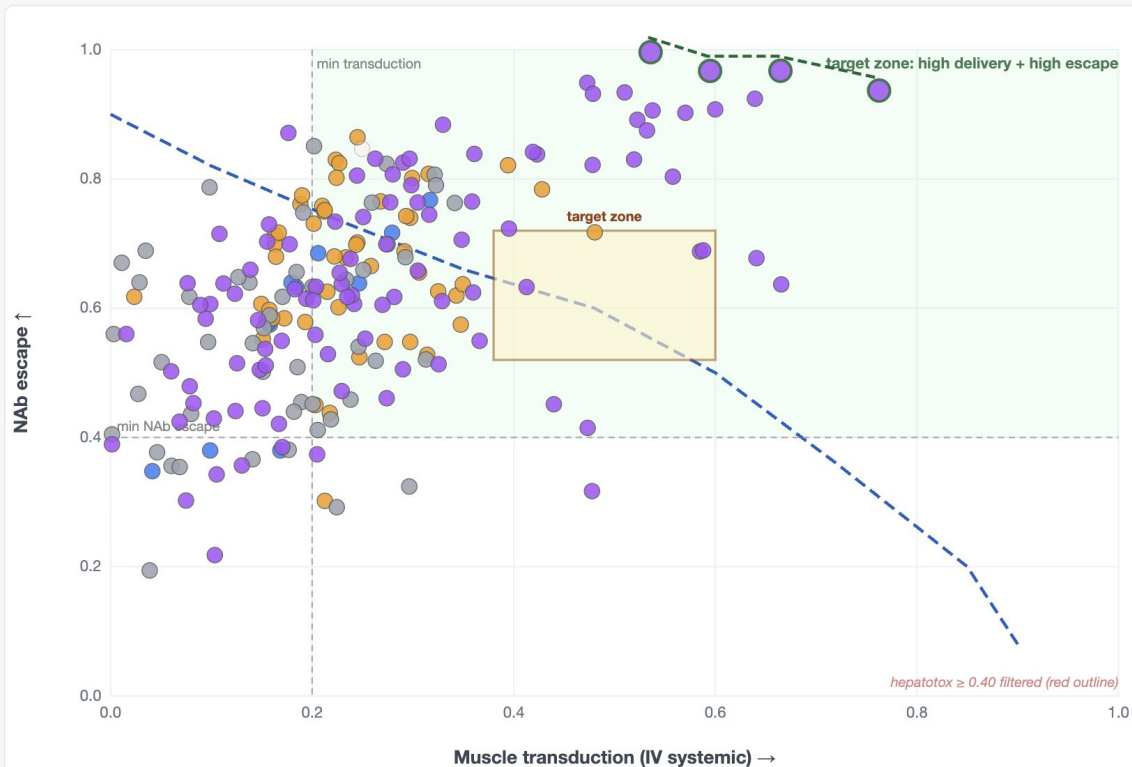
RL policy

random baseline

seed

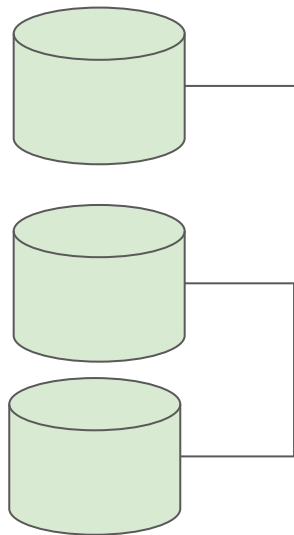
extended pool (+100)

★ frontier only

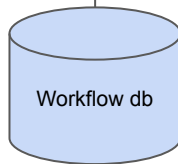
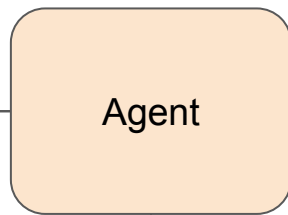


Reasoning-oriented architecture

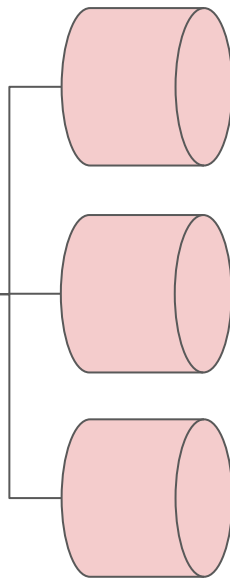
Pre-computed data



MCP interfaces



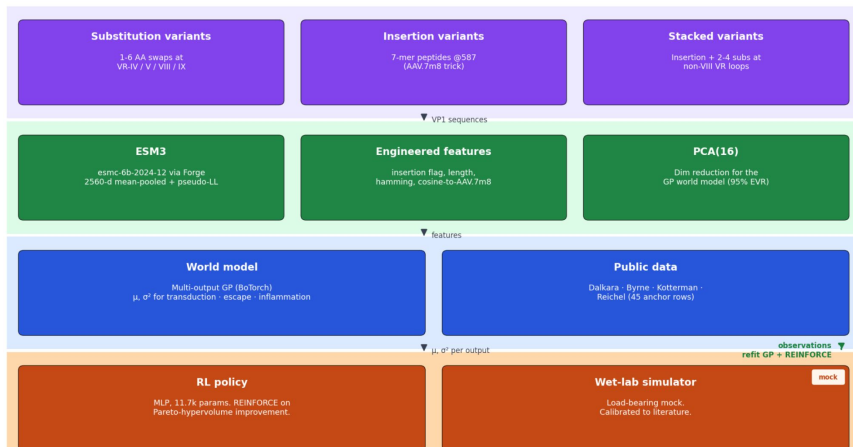
Low-latency models + tools



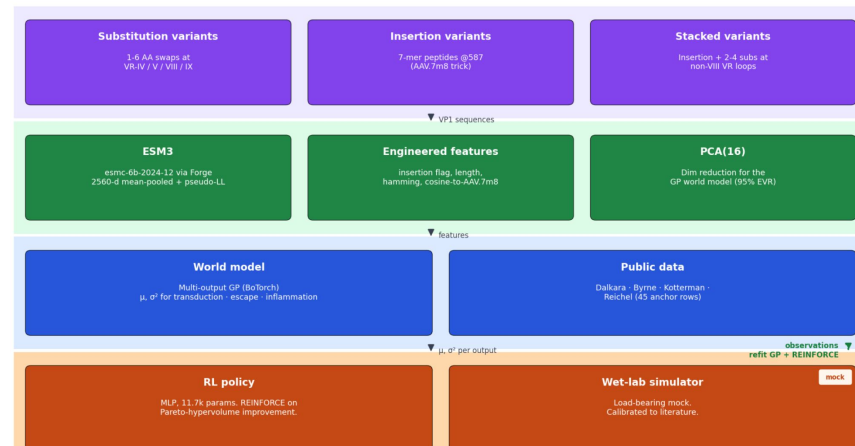
Read-only
architecture
unlocks ability
scale exploration

This architecture powers 2 world models

Mechanistic hypothesis world model



AAV therapeutic world model



Generator of synthetic clinical +
AAV-assay data

Feedback on the platform from my collaborators

Most large-scale biobanks were designed for association mapping rather than mechanistic validation, safety inference, or clinical strategy development. As a result, pharmaceutical companies often lack access to deeply phenotyped, causally enriched human datasets capable of answering practical development questions such as:

- Which targets are safe to inhibit?
- Which variants modify disease progression rather than risk?
- Which patient subgroups are most likely to respond to intervention?
- How do we therapeutically intervene?

This rare disease-anchored translational infrastructure platform is optimized to close this gap.

Why I built this

Highly motivated
patient and
caregiver
populations

Limited numbers
of clinicians and
researchers

Few, if any,
therapeutic
options

I'm working with rare disease researchers focused on rare diseases affecting children. The idea is for this platform to be deployed to aid in their research. I'd like to see it used more broadly

Next 90 days

QC this
framework
against well
characterized
targets/diseases
(PCSK9)

Integrate new
analytic
workflows into
this framework

Building a
collaboration
with sample
management
orgs

Let's collaborate!

Are you a
researcher and
see value?

Do you want to
help build or
extend this?

Do you want to
fund the creation
of this platform?



AI for Science Build Fellows (Spring 2026)

▼ 1 more property



Fellow Profiles

Gallery view



Vivek



Mohamed



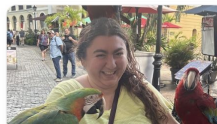
Dale



Natalie



Akhil



julia



Sameer



Closing the loop from pathogenic variant to gene therapy

Scaling our ability to explore the pareto frontier of patient-specific mechanistic hypotheses and therapeutic variants